

# RNA counts, normalization and defensible expression claims

When is a bigger count not evidence of higher expression?

# By the end, you should be able to...

Interpret a genes-by-samples count matrix and sample metadata.

Calculate a toy TPM correctly and distinguish it from count-model input.

Read a differential-expression summary without claiming protein function.

# Reconnect with the biology

A gene may produce several transcripts; counts depend on what features and assignment rules were used.

**Think silently -> explain to a neighbour -> check the measurement.**

# From RNA to a count matrix

**01**

Reads are assigned to genes or transcripts using a stated reference and rule.

>

**02**

Rows represent features; columns represent samples in our bulk RNA example.

>

**03**

Sample metadata must match columns by identifier, not accidental order.

# Counts depend on more than biology

Library size changes how many molecules are sampled.

Feature length and composition affect some counting summaries.

Biological differences and technical biases require an explicit model.

# Normalization is not one universal operation

**Relative abundance display**

**Count-based inference**

Library-size scaling addresses a sampling-depth component.

TPM divides by length and rescales within each sample in our toy example.

For differential expression, use a method with its specified input and assumptions.

# Toy TPM: length first, sum second

$$\text{TPM}_i = (\text{count}_i / \text{length\_kb}_i) / \text{sum}(\text{rates}) \times 1,000,000$$

State the units and the assumptions.

Divide each feature count by its length in kilobases.

Sum these length-adjusted rates within the sample.

Divide each rate by that sum and multiply by one million.

# An invariant can catch the wrong calculation

| Feature | Count | Length kb | Rate |
|---------|-------|-----------|------|
| A       | 100   | 1         | 100  |
| B       | 200   | 2         | 100  |

TPM should sum to one million within a non-empty sample.

But sum-to-one-million alone does not prove the calculation is correct.

Also compare an equal-abundance, unequal-length control.

# DESeq2 is not a t-test on TPM

A count model uses sample-level replication and its own normalization.

Raw count-matrix workflows should not be fed TPM values.

Approved transcript-quantifier import workflows are a separate route; follow their documentation.

# Fold change and evidence are separate axes

| Gene  | log2 change | Adjusted p |
|-------|-------------|------------|
| Toy A | 2.0         | 0.30       |
| Toy B | 0.2         | 0.001      |

log2 fold change describes the estimated direction and magnitude.

Adjusted p-values address a family of tests under assumptions.

Large effects can be uncertain; tiny effects can be statistically supported.

# A result table is not a mechanism

**01**

RNA changes can reflect cell composition or regulation.

>

**02**

Enrichment depends on the tested gene universe and annotation choices.

>

**03**

State what additional protein, functional or replication evidence is needed.

## Worked example: equal rates, unequal counts

Two toy features: A has 100 counts and length 1 kb; B has 200 counts and length 2 kb.

**Predict the result and one limitation before calculating.**

# Work through the reasoning

Rates are  $100 / 1 = 100$  and  $200 / 2 = 100$ .

Their rate sum is 200.

TPM values are 500,000 each; raw counts differ while length-adjusted rates are equal.

## Pause: make a defensible claim

An analyst rounds TPM and passes it to a raw-count DESeq2 workflow. Why is this not repaired by rounding?

**First answer alone. Then compare the evidence, not just the conclusion.**

# A defensible response

Rounding cannot reconstruct the original count sampling information or the intended count model. Use the documented raw-count or supported import workflow.

# Today's practical

W07: repair the TPM order-of-operations bug; then align sample metadata in a real two-gene pasilla excerpt. Do not perform DE on that excerpt.

| Minutes | Action                |
|---------|-----------------------|
| 0-10    | Predict + set up      |
| 10-30   | Run + inspect + check |
| 30-40   | Explain + compare     |
| 40-45   | Save + submit / exit  |

# Five terms to use precisely

1 count matrix

2 metadata

3 normalization

4 TPM

5 fold change

# Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

# Go further only after the core is clear

Optional: inspect median-of-ratios normalization and effective lengths; no dispersion derivation is assessed.

**Not required for the core grade. No advanced model training is expected.**

# Read the evidence, not only the slide

S09 Love, Huber and Anders (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2.

S10 DESeq2: Analyzing RNA-seq data with DESeq2, release vignette.

S25 Huber and Reyes. pasilla data package; Brooks et al. (2011) experiment.